The Saturation Theorem: Axiomatic Constraints on Quantum Gravity from Cosmological Relaxation

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We formulate four minimal axioms – finite geometric capacity, cooperative reinforcement, monotone cosmic control, and renormalization-group composition – as candidate macroscopic constraints for any quantum gravity theory whose cosmological limit reproduces general relativity with a bounded relaxation mechanism. These axioms imply an Abel-coordinate description of one-dimensional saturation dynamics. A hyperbolic tangent profile follows once the saturation boundary is fixed by an additional projective boundary-quotient normalization: the ratio $R(x) = (1+x)/(1-x)$ must multiply under composition. Without that normalization, smooth boundary regularity alone leaves a family of smooth hyperbolic boundary collars, for example Abel coordinates of the form $\operatorname{arctanh}(x) + \varepsilon x$. The corrected *Saturation Theorem* is therefore a projective-collar normal-form theorem: $S(g) = S_{\max} \tanh(\alpha\, u(g))$ is selected by scale coherence plus the projective boundary quotient, not by bare smoothness alone. Major quantum gravity programs – Loop Quantum Gravity, asymptotic safety, string/holographic approaches, causal set theory, and noncommutative geometry – naturally motivate A–D when restricted to cosmological observables, while D' remains a separate diagnostic normalization to be physically derived. The result elevates the tanh saturation ODE $dX/da = k(1 - X^2)$ of the Curvature Relaxation Model from a phenomenological fit to a canonical projective normal form of capacity-limited cooperative relaxation, while leaving open which UV completion selects k, Φ_0, and the projective quotient physically.

I. INTRODUCTION

A. The problem

The unification of quantum mechanics and general relativity remains the central open problem of theoretical physics. After decades of effort, several major programs have produced deep structural insights without converging on a single theory:

- **Effective field theory (EFT) of gravity** computes reliable quantum corrections at low energies but provides no UV completion [5].
- **Asymptotic safety** seeks a UV fixed point of the gravitational renormalization group (RG) flow, constraining the theory without new degrees of freedom [8–10].
- **String theory** provides a UV-finite framework containing gravity automatically, but faces the landscape/selection problem [11].
- **Loop Quantum Gravity (LQG)** achieves background independence with discrete geometry operators, but struggles with dynamics and the classical limit [12, 13].
- **Holography/AdS-CFT** offers a nonperturbative definition of quantum gravity in anti-de Sitter space, with unclear extension to realistic cosmology [14].
- **Causal set theory** posits discrete causal structure as fundamental, with the continuum as emergent [15, 16].
- **Noncommutative geometry** reformulates geometry as spectral data, naturally bridging operators and spacetime [17].

Orthogonal to these UV-completion programs, the *Swampland conjectures* [30] constrain which low-energy theories can be consistently embedded in quantum gravity, while the *EFT of Dark Energy* [31] parametrizes deviations from ΛCDM at the perturbative level. Neither framework, however, determines the *functional form* of the cosmological saturation profile.

A striking observation is that when any of the UV-completion programs listed above is applied to cosmology, the resulting effective dynamics generically exhibits *saturation behavior*: a bounded, cooperative approach to a maximal geometric configuration. This paper asks whether this universality is accidental or inevitable.

B. The CRM as empirical anchor

The Curvature Relaxation Model (CRM) [1–4] provides a concrete, empirically tested cosmological framework in which saturation dynamics plays the central role. The model replaces the cosmological constant

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Λ and particulate dark matter with geometric curvature return, achieving $\Delta\chi^2 = -5.5$ vs. ΛCDM on joint SN+CMB+BAO data with 6 parameters and zero Early Dark Energy [2]. The entire framework derives from a single dynamical equation:

$$\frac{dX}{da} = k(1 - X^2)\,, \tag{1}$$

where $X = \Omega_\Phi/\Phi_0$ is the normalized curvature return potential and a is the scale factor. The solution $X(a) = \tanh(k(a - a_{\rm trans}))$ provides late-time acceleration ("dark energy") in its saturated phase and gravitational scaffolding ("dark matter") in its unsaturated phase [2].

Paper III [3] derives this ODE from the effective Lagrangian $\mathcal{L}_{\rm CRM} = R/(16\pi G) + \gamma\mathcal{F}(T/\rho)R^2 + \frac{1}{2}(\partial\phi)^2 - V_{\rm PT}(\phi)$, where $V_{\rm PT}(\phi) = V_0/\cosh^2(\phi/\phi_0)$ is a Pöschl-Teller potential. Paper III also surveys five UV-completion candidates (LQG, Finsler geometry, information-theoretic spacetime, causal sets, quantum error correction), observing that all produce ODE structures of the form $dX/dt \propto (1 - X^2)$.

The present paper elevates this observation from *structural analogy* to a conditional normal-form theorem: the four physically motivated axioms give an Abel-coordinate saturation law, and the additional projective boundary-quotient normalization selects the tanh representative.

C. Scope and structure

The central result is the *Saturation Theorem* (Theorem 1): Axioms A–D, together with signed antisymmetry B' and the interior group form $D^\pm$, reduce the effective cosmological response to an Abel coordinate, while the projective boundary-quotient normalization D' uniquely selects the tanh normal form up to reparametrization. The paper is organized as follows:

- Section II: Formulation of the four axioms.
- Section III: Statement and proof of the Saturation Theorem.
- Section IV: Exclusion of alternative saturation profiles.
- Section V: Verification of the axioms in major QG programs.
- Section VI: Ontological implications: spacetime as an ordered phase.
- Section VII: Observational consequences and open questions.
- Section VIII: Discussion and outlook.

II. THE FOUR AXIOMS

We formulate the axioms in terms of an effective response function $S_T(g)$, where T denotes a quantum gravity theory and g is an effective coupling/curvature/backreaction parameter measuring the "distance" from the UV to the IR. The function $S_T(g)$ encodes how the theory's geometric output (curvature, expansion rate, geometric order) responds to g.

A. Axiom A: Finite Geometric Capacity

Axiom 1 (Finite Geometric Capacity)**.** *Per Hubble patch, there exists a finite maximum geometric order $C_{\rm max} < \infty$. Equivalently, the effective response function $S_T(g)$ is globally bounded:*

$$0 \le S_T(g) < S_{\rm max} < \infty \quad \forall\, g \ge 0\,. \tag{2}$$

Physical content. This axiom excludes infinite curvature, infinite information density, and UV divergences. It is extremely weak: it holds in LQG (finite area quanta [12]), in holography (Bekenstein bound [18]), in quantum error-correcting spacetime (finite code capacity [19]), in causal set theory (finite density [15]), and in asymptotic safety (finite effective action at the UV fixed point [9]).

Without finite capacity, quantum gravity is by definition unstable: unbounded geometric configurations correspond to singularities, which every QG program aims to resolve.

Connection to CRM. The saturation amplitude Φ_0 in the CRM [1] is the physical realization of $S_{\rm max}$. The Pöschl-Teller potential $V(\phi) = V_0/\cosh^2(\phi/\phi_0)$ enforces $\Omega_\Phi \le \Phi_0$ dynamically [3].

B. Axiom B: Cooperative Reinforcement

Axiom 2 (Cooperative Reinforcement)**.** *Existing geometric order facilitates further ordering. The effective response function $S_T(g)$ is smooth (C^∞) for $g \ge 0$, with a local expansion at $g = 0$:*

$$S_T(g) = c_1 g + c_3 g^3 + \mathcal{O}(g^5)\,, \quad c_1 > 0\,. \tag{3}$$

The odd-power structure reflects the catalytic nature of the response: the rate of geometric ordering is proportional to both the existing ordered fraction and the remaining capacity.

Physical content. This axiom encodes the cooperative, self-reinforcing character of geometric ordering. It is the gravitational analog of ferromagnetic ordering, where aligned spins facilitate further alignment. Formally, the condition $c_1 > 0$ ensures that the response is positive (ordering, not disordering) for small perturbations, while the odd-power structure is consistent with the antisymmetric extension (Axiom B').

Without cooperative reinforcement, there would be no classical spacetime: a linear or anti-cooperative response cannot produce the long-lived, large-scale geometric structures observed in the universe. The cooperative character is analogous to spontaneous symmetry breaking in condensed matter: local ordering seeds further ordering, driving the system toward a globally ordered state (saturation).

Connection to CRM. The saturation coupling k in the CRM ODE (1) is c_1. The game-theoretic equilibrium of Paper I [1] provides the thermodynamic motivation: the null space's concentration gradient G drives cooperative ordering until the spacetime bubble saturates.

A companion condition extends the domain of S_T to negative arguments, enabling the group-theoretic classification of Step 2 in the proof:

Axiom 3 (Antisymmetric Extension (B'))**.** *The response function S_T admits a smooth antisymmetric extension to all $g \in \mathbb{R}$:*

$$S_T(-g) = -S_T(g) \quad \forall\, g\,. \tag{4}$$

Physical content. The antisymmetry has three complementary motivations:

(i) Sign covariance of the response. The parameter g is not a "distance" (which would be non-negative by definition) but a *signed deviation* from a symmetric reference state. In the CRM, g parametrizes the curvature perturbation relative to the de Sitter background; positive g corresponds to over-dense perturbations that enhance geometric ordering, and negative g to under-dense regions that reduce it. Sign covariance ($S_T(-g) = -S_T(g)$) then states that the response function treats over- and under-dense regions symmetrically in magnitude – a natural physical requirement absent any symmetry-breaking mechanism in the response itself. We emphasize that Axiom B' is *not* a claim about cosmological time-reversal symmetry (which is broken by the expansion); it is a constraint on the functional form of the response under $g \to -g$.

(ii) Precedents in physics. Sign covariance of response functions is ubiquitous: in electrodynamics, the polarization $P(-E) = -P(E)$ for centrosymmetric media; in magnetism, $M(-H) = -M(H)$ for the paramagnetic/ferromagnetic response; in the renormalization group, beta functions of dimensionless couplings are odd functions of the coupling deviation from a fixed point. The antisymmetry of Axiom B' is the gravitational analog of these well-established symmetries.

(iii) Mathematical role. Together with the odd-power structure of Axiom B, B' extends the composition law from the half-line $[0, S_{\max})$ to a full group operation on $(-S_{\max}, S_{\max})$, which is essential for the Abel-equation classification. Without B', only a semigroup structure is available, and the Abel-representation argument of Step 2 does not apply in the same form. We note that this is an explicit assumption, not a derived consequence; its relaxation leads to alternative composition structures (e.g., $C(x, y) = x + y - xy$ on $[0, 1)$), which is why it appears as a separate axiom rather than being subsumed into Axiom B.

C. Axiom C: Monotone Cosmic Control

Axiom 4 (Monotone Cosmic Control)**.** *The effective response function $S_T(g)$ is monotonically increasing, with saturation at infinity:*

$$S_T'(g) \geq 0\,, \quad \lim_{g\to\infty} S_T(g) = S_{\max}\,. \tag{5}$$

There exists exactly one relevant direction (control parameter) governing the transition from low to high geometric order.

Physical content. This excludes oscillating, multi-fixed-point, or chaotic geometric dynamics. It guarantees:

- A unique time direction (thermodynamic arrow).
- No "bouncing" between geometric phases.
- Stable expansion toward a de Sitter attractor.

In the CRM, the control parameter is the scale factor a; in RG language, it corresponds to the single relevant direction flowing from the UV fixed point to the IR [10]. Axiom C is violated by cyclic cosmologies and by models with multiple stable vacua – both of which lack a unique thermodynamic arrow. We note that Axiom C is a restriction to the *cosmological sector* with a single dominant transition: it does not address the string landscape (where tunneling between metastable vacua produces non-monotone trajectories) or stochastic inflation (where local fluctuations break monotonicity). These scenarios involve multiple effective control parameters or stochastic dynamics, placing them outside the scope of the single-field, deterministic axiom system. The theorem should be read as follows: A–D, with B' and the interior group form $D^{\pm}$, give an Abel/collar class for a single deterministic saturation channel; adding D' selects the projective tanh representative.

Connection to CRM. The monotonicity of $\Omega_\Phi(a)$ is a direct consequence of the ODE (1): since $1 - X^2 > 0$ for $|X| < 1$, the solution is strictly increasing.

Remark 1 (Axiom interdependence)**.** *As shown in Corollary 1, monotonicity (Axiom C) follows once Axiom D is used in its interior-group form $D^{\pm}$. We nevertheless retain Axiom C as an explicit statement for physical transparency: it connects directly to the thermodynamic arrow of time and makes the exclusion of oscillating/chaotic cosmologies visible at the axiom level. The logical content of the axiom system is thus: A, B/B', and the strengthened composition law $D^{\pm}$ imply C; without that interior-group condition, C remains an explicit axiom.*

D. Axiom D: Renormalization-Group Composition

Axiom 5 (Renormalization-Group Composition)**.** *There exists a scale composition $g \mapsto g_1 \oplus g_2$ ("two RG steps in succession") such that the response function is compositionally coherent:*

$$S_T(g_1 \oplus g_2) = \mathcal{C}(S_T(g_1),\, S_T(g_2)) \;, \tag{6}$$

where $\mathcal{C}$ is a smooth, commutative, associative composition with neutral element 0 and absorbing boundary $S_{\max}$:

$$\mathcal{C}(S, 0) = S \;, \quad \mathcal{C}(S, S_{\max}) = S_{\max} \,. \tag{7}$$

Furthermore, $\mathcal{C}$ extends smoothly (i.e., C^∞) to the closed square $[0, S_{\max}]^2$. In the theorem below, D is used in its antisymmetric interior-group form $D^\pm$: after normalization by $S_{\max}$ and extension by B', the restriction $C : (-1,1)^2 \to (-1,1)$ is cancellative and strictly monotone in each argument, and each translation $y \mapsto C(x,y)$ is a smooth diffeomorphism with inverse $y \mapsto C(-x,y)$. This is an additional one-dimensional deterministic-channel assumption, not a consequence of the bare positive-face semigroup law alone.

Remark 2 (Boundary-collar guardrail)**.** *The smooth-extension clause in Axiom D is not, by itself, used below as a uniqueness claim for* arctanh*. The Abel coordinate*

$$\phi_\varepsilon(x) = \operatorname{arctanh}(x) + \varepsilon x$$

induces a smooth associative composition on the open interval and gives a smooth positive boundary collar via

$$q_\varepsilon(x) = e^{-2\phi_\varepsilon(x)} = \frac{1-x}{1+x} e^{-2\varepsilon x}.$$

Thus bare smoothness of the positive saturation face selects a hyperbolic collar class, not a unique projective coordinate. The tanh *normal form becomes a theorem only after the projective boundary quotient is fixed explicitly.*

Axiom 6 (Projective Boundary-Quotient Normalization (D'))**.** *For the normalized response $x = S/S_{\max} \in (-1,1)$, the projective boundary quotient*

$$R(x) := \frac{1+x}{1-x}$$

is multiplicative under composition:

$$R(C(x,y)) = R(x)R(y). \tag{8}$$

Equivalently, the ratio of distances to the two saturation faces is the additive work/rapidity coordinate of the one-dimensional response channel.

Physical status of D'. D' is not derived here from LQG, asymptotic safety, string theory, causal sets, CDT, or noncommutative geometry. It is a projective response-balance assumption: the physically composed quantity is the ratio of realized order to remaining capacity, analogous to a log-odds or rapidity coordinate. For any candidate UV program, the diagnostic defect is

$$\Delta_{D'}(x,y) := \log R(C(x,y)) - \log R(x) - \log R(y),$$

which must vanish for the exact projective tanh representative. Establishing $\Delta_{D'} = 0$ from a microscopic theory remains an open physical task.

Physical content of D. Axiom D is the deepest composition axiom. It states that the geometric response must be *self-consistent under coarse-graining*: observing the universe at two successive scales and combining the results must give the same answer as observing at the combined scale directly. This is the defining property of a renormalization group.

The composition law (6) encodes the requirement that the theory be *scale-coherent* in the following precise sense: the response function S_T is a homomorphism from the additive structure of the control parameter to the composition algebra of responses; i.e., no preferred scale exists at which this homomorphism property breaks down or changes character. The absorbing boundary condition $\mathcal{C}(S, S_{\max}) = S_{\max}$ reflects the fact that fully saturated geometry cannot be further ordered – it is a fixed point.

The C^∞-extension requirement has a direct physical interpretation: it demands that the composition law remains well-behaved *even near saturation.* In physical terms, this means that the approach to the geometric capacity limit is smooth and does not introduce new divergences or critical exponents. Three physical arguments support this requirement:

(a) No new phase transition at saturation. A composition that develops singular derivatives at the boundary would signal a phase transition *within* the saturation process itself – i.e., the saturation mechanism would break down precisely where it is needed most. The smoothness condition excludes such pathological behavior. This is distinct from phase transitions at the *onset* of saturation (which are permitted and correspond to the CRM's transition epoch a_{trans}); the boundary regularity concerns the *fully saturated* regime.

(b) Distinction from critical phenomena. While critical exponents and divergent susceptibilities characterize phase transitions at finite temperature (the Curie point in ferromagnets, T_c in superfluids), the saturation boundary $S \to S_{\max}$ is an *asymptotic* regime ($g \to \infty$), not a critical point. In the ferromagnetic analogy, it corresponds to $T \to 0$ (complete alignment), not to $T \to T_c$ (critical fluctuations). The zero-temperature limit of the magnetization *is* smooth ($M \to M_s$ exponentially in mean-field theory and as a power law in low T, but without divergent derivatives of the *composition* of magnetizations). The C^∞ requirement is therefore physically appropriate for the saturation boundary.

(c) Analyticity of the vacuum state. In quantum field theory, the vacuum is an analytic state: all correlation

functions at the vacuum are smooth. Since full saturation ($S = S_{\max}$) represents the geometric analog of the vacuum (the maximally ordered state), the composition of responses near this state should inherit the smoothness of the vacuum.

Connection to CRM. In the CRM, the composition law emerges from the thermodynamic structure: the curvature return potential Ω_Φ at scale $a_1 \cdot a_2$ is determined by the potentials at a_1 and a_2 through the saturation ODE's group property. The tanh function satisfies $\tanh(u_1+u_2) = (\tanh u_1+\tanh u_2)/(1+\tanh u_1 \tanh u_2)$, which is precisely a composition law of the required form with $\mathcal{C}(x, y) = (x + y)/(1 + xy/S_{\max}^2)$.

III. THE SATURATION THEOREM

A. Statement

Theorem 1 (Saturation Theorem, projective-collar form)**.** *Let T be a theory whose effective response function $S_T(g)$ satisfies Axioms A–D, the antisymmetry condition B', and the interior-group form $D^\pm$ described in Axiom D. Then the normalized response $\sigma(g) = S_T(g)/S_{\max}$ admits an odd Abel coordinate $\phi : (-1, 1) \to \mathbb{R}$ such that*

$$\phi(C(x, y)) = \phi(x) + \phi(y).$$

If, in addition, the projective boundary-quotient normalization D' holds, then there exist constants $\alpha > 0$, $S_{\max} > 0$, and a strictly monotone reparametrization $u = u(g)$ with $u(0) = 0$, such that

$$S_T(g) = S_{\max} \tanh(\alpha\, u(g)) \ . \tag{9}$$

The composition law is then determined as the relativistic velocity addition $\mathcal{C}(x, y) = (x+y)/(1+xy)$, up to rescaling by $S_{\max}$. Without D', these hypotheses determine the Abel/collar structure but do not exclude all smooth non-projective collars.

B. Key Lemmas

The uniqueness of arctanh rests on a projective quotient, not on smoothness alone.

Lemma 1 (Projective quotient selects arctanh)**.** *Let C be a continuous, strictly monotone, commutative and associative group law on $(-1, 1)$ with neutral element 0. Suppose C has an Abel coordinate ϕ,*

$$\phi(C(x, y)) = \phi(x) + \phi(y),$$

and satisfies the projective quotient law (8). Then there exists $c > 0$ such that

$$\phi(x) = c \operatorname{arctanh}(x), \qquad C(x, y) = \frac{x + y}{1 + xy}.$$

Proof. Set $L(x) := \frac{1}{2}\log((1 + x)/(1 - x)) = \operatorname{arctanh}(x)$. By D',

$$L(C(x, y)) = L(x) + L(y).$$

Thus $F(t) := \phi(\tanh t)$ is a continuous additive function of $t = L(x)$, so $F(t) = ct$ for some $c > 0$. Hence $\phi(x) = cL(x) = c \operatorname{arctanh}(x)$, and solving $L(C) = L(x) + L(y)$ gives $C(x, y) = (x + y)/(1 + xy)$. □

Remark 3 (Why D' is necessary)**.** *The family $\phi_\varepsilon(x) = \operatorname{arctanh}(x)+\varepsilon x$ is odd, smooth and strictly increasing for small ε, diverges at the boundary, and induces a smooth positive saturation collar through $q_\varepsilon = e^{-2\phi_\varepsilon}$. Hence the old claim that C^∞ boundary regularity alone excludes all non-*arctanh *Abel coordinates is too strong. The corrected theorem uses D' as the mathematical and physical normalization that removes this boundary-gauge freedom.*

C. Proof of the Saturation Theorem

The proof proceeds in three steps.

Step 1: Derivation of the functional equation. Define the normalized response $\sigma(g) = S_T(g)/S_{\max} \in [0, 1)$. Axioms A and C guarantee that σ is bounded and monotonically increasing. Axiom D [Eq. (6)] requires the existence of a composition:

$$\sigma(g_1 \oplus g_2) = C(\sigma(g_1), \sigma(g_2))\,, \tag{10}$$

where $C : [0, 1) \times [0, 1) \to [0, 1)$ is commutative, associative, has neutral element 0, and absorbing boundary 1.

Step 2: Reduction to Abel's equation. By Axiom B, σ has the local expansion (3) with $c_1 > 0$. Axiom B' [Eq. (4)] provides the odd extension $\sigma(-g) = -\sigma(g)$. Together with the interior-group form $D^\pm$ of Axiom D, this gives a smooth cancellative group operation on $(-1, 1)$ with absorbing boundaries at ±1 treated only as limiting faces.

Under $D^\pm$, the associativity condition $C(C(x, y), z) = C(x, C(y, z))$ and commutativity define a *one-parameter continuous group law* on $(-1, 1)$ (the Taylor expansion at the origin defines a one-dimensional commutative formal group law in the sense of [21]; here the connection is that any smooth, associative, commutative binary operation on a neighborhood of $0 \in \mathbb{R}$ with neutral element 0 is, by Taylor expansion, a formal group law over $\mathbb{R}$, and conversely, Aczél's analytic theorem applies to the convergent realization of this formal structure on the interval $(-1, 1)$). We apply Aczél's representation theorem [20] to the *open interval* $(-1, 1)$, where C is smooth, strictly monotone in each argument, and the neutral element 0 lies in the interior. The absorbing boundary conditions $C(x, \pm1) = \pm1$ are *not* part of the domain on which Aczél's theorem is applied; they are treated as limiting behavior. Specifically, Aczél's theorem yields a continuous, strictly monotone function $\phi : (-1, 1) \to \mathbb{R}$ with

$\phi(0) = 0$. The absorbing boundary conditions then imply $\phi(x) \to +\infty$ as $x \to 1^-$ and $\phi(x) \to -\infty$ as $x \to -1^+$ (since $C(x,y) \to 1$ as $y \to 1$ forces $\phi(x) + \phi(y) \to \infty$, which requires $\phi(y) \to \infty$):

$$\phi(C(x,y)) = \phi(x) + \phi(y)\,. \tag{11}$$

This is Abel's functional equation. Aczél's theorem yields ϕ continuous; since C is smooth (Axiom D) and $\phi'(0) = c_1 > 0$ (Axiom B), differentiating (11) shows that ϕ inherits the C^∞ regularity of C on $(-1,1)$. The function ϕ is determined up to a positive multiplicative constant, and antisymmetry (Axiom B') lets it be chosen odd. At this stage Axioms A–D give an Abel/collar theorem, not yet $\phi = \operatorname{arctanh}$.

Step 2b: Projective boundary normalization. If the additional normalization D' holds, Lemma 1 gives $\phi = c\operatorname{arctanh}$. This yields the composition law:

$$C(x,y) = \frac{x+y}{1+x\,y}\,, \tag{12}$$

which is the relativistic velocity addition formula. The uniqueness is therefore projective: it is uniqueness after fixing the boundary quotient $R = (1+x)/(1-x)$, not uniqueness from bare smoothness of all Abel collars.

Step 3: Inversion to tanh. Since $\phi(\sigma(g)) = c\operatorname{arctanh}(\sigma(g))$ must be an additive function of the control parameter (i.e., $\phi(\sigma(g)) = \alpha\,u(g)$ for a reparametrization u and constant $\alpha > 0$), we obtain:

$$\sigma(g) = \tanh(\alpha\,u(g))\,, \tag{13}$$

and therefore $S_T(g) = S_{\max}\tanh(\alpha\,u(g))$, after absorbing the positive constant c into αu. When Axioms A–D, B', $D^\pm$, and D' hold exactly, the result is exact. Approximate axiom satisfaction (e.g., breakdown of the composition law at very small g where standard GR dominates) would introduce corrections; a formal stability analysis of such perturbations is an open problem. □

Corollary 1 (Monotonicity from the interior group law)**.** *Under the interior-group form $D^\pm$ of Axiom D, Axiom C (monotone cosmic control) follows from Axioms A, B, B', and D. That is, any response function satisfying $A + B + B' + D^\pm$ is automatically strictly monotonically increasing on the one-dimensional channel.*

Proof. Differentiate $\sigma(g+h) = C(\sigma(g), \sigma(h))$ with respect to h at $h = 0$:

$$\sigma'(g) = \partial_2 C(\sigma(g), 0)\cdot\sigma'(0)\,.$$

Define $a(x) := \partial_2 C(x,0)$. By $D^\pm$, for each fixed $x \in (-1,1)$ the map $y \mapsto C(x,y)$ is a C^∞ diffeomorphism of $(-1,1)$ (its inverse is $y \mapsto C(-x,y)$ from the group inverse). Therefore $\partial_2 C(x,0) = a(x) \neq 0$ for all $x \in (-1,1)$. Since $(-1,1)$ is connected and $a(0) = \partial_2 C(0,0) = 1 > 0$ (from $C(0,y) = y$), continuity implies $a(x) > 0$ for all x. With $\sigma'(0) = c_1 > 0$ (Axiom B), we obtain $\sigma'(g) = a(\sigma(g))\cdot c_1 > 0$ for all g. □

Remark 4. *This means the axiom system can be reduced to A, B/B', and $D^\pm$ plus the derived consequence C whenever the one-dimensional interior group law is physically justified. We retain Axiom C as a separate statement for physical transparency: it makes the monotonicity assumption explicit and connects directly to the thermodynamic arrow of time. Without $D^\pm$, it should be read as an independent single-channel assumption.*

Proposition 1 (Axiom Independence)**.** *The axioms are logically independent: for each, there exists a function satisfying the others but violating the chosen one.*

(1) Violates A, satisfies B/B'–D: $\sigma(g) = g$ (unbounded linear response; $C(x,y) = x+y$, associative and commutative).

(2) Violates B', satisfies A,C,D: $\sigma(g) = 1 - e^{-g}$ for $g \geq 0$, with $C(x,y) = x + y - xy$ (probabilistic or); no antisymmetric extension.

(3) Violates C, satisfies A,B/B' but not $D^\pm$: $\sigma(g) = \tanh(g)\cdot\cos(\epsilon g)$ for small $\epsilon > 0$; bounded, antisymmetric, smooth with $\sigma'(0) > 0$, but non-monotone for $g \gg 1$. This example shows that C is independent of $\{A, B, B'\}$ alone. However, by Corollary 1, C is not independent of $\{A, B, B', D^\pm\}$ jointly – it follows as a derived consequence under the interior group law. Thus C is retained for physical clarity and to make the single-channel arrow explicit.

(4) Violates D, satisfies A–C,B': $\sigma(g) = (2/\pi)\arctan(g)$; bounded, monotone, antisymmetric, but its induced composition (via $\phi = \tan(\pi x/2)$) develops divergent second derivatives ($\sim \epsilon^{-1}$) at the boundary of $[-1,1]^2$, failing the C^∞-extension requirement of Axiom D. Similarly, $\sigma(g) = g/\sqrt{1+g^2}$ (with $\phi(x) = x/\sqrt{1-x^2}$) has divergent second derivatives at the boundary ($\sim \epsilon^{-1/2}$).

D. Interpretation

The theorem states that tanh is not a mere *model choice*, but the *canonical projective representative* of a one-dimensional Abel saturation channel. The key mathematical insight is two-step: Axiom D forces the response function to be the inverse of an Abel coordinate, and D' fixes the otherwise free boundary collar by making the projective quotient $R = (1+x)/(1-x)$ multiplicative.

Physically, the theorem says:

> *Any quantum gravity theory that is scale-coherent, cooperative, bounded, monotone, and whose saturation boundary is normalized by the projective quotient must produce cosmological saturation dynamics of the* tanh *type. Without that quotient normalization, the theorem gives an Abel/collar class rather than a unique profile.*

The reparametrization $u(g)$ absorbs the microscopic details of the theory: different UV completions correspond to different functions $u(g)$. The common $\tanh(\alpha\,u)$

form is forced only after the projective boundary ratio has been identified as the physical composition variable.

IV. EXCLUSION OF ALTERNATIVE PROFILES

Corollary 2 (Exclusion of Alternative Saturation Profiles)**.** *None of the following profiles can simultaneously satisfy the theorem's structural hypotheses plus the projective normalization D', except when reducible to the* tanh *form via admissible reparametrization $u(g)$:*

(i) ***Logistic profile:*** $S_{\max}/(1+e^{-\beta u})$.

(ii) ***Rational profile:*** $S_{\max}\, u/(1+u)$.

(iii) ***Arctangent profile:*** $S_{\max}\,(2/\pi)\arctan(\beta u)$.

(iv) ***Hard saturation:*** $S_{\max}\,\min(u,1)$.

Proof. Each alternative fails a specific axiom:

(i) Logistic. The logistic function $L(u) = 1/(1+e^{-\beta u})$ has $L(0) = 1/2 \neq 0$, violating the neutral element condition. The shifted logistic $\tilde{L}(u) = (L(u) - 1/2)\cdot 2$ yields $\tilde{L}(u) = \tanh(\beta u/2)$ – precisely the tanh form. Therefore, any "logistic" model that satisfies Axiom D is actually tanh in disguise.

(ii) Rational. The function $R(u) = u/(1+u)$ has Abel transform $\phi(x) = x/(1-x)$, yielding $C(x,y) = \phi^{-1}(\phi(x)+\phi(y)) = (x+y-2xy)/(1-xy)$. For small x, y, this gives $C(x,y) \approx x+y-2xy$. The quadratic cross-term $-2xy$ is incompatible with the antisymmetry condition (Axiom B'): an antisymmetric σ implies that the induced composition must satisfy $C(-x,-y) = -C(x,y)$, but $(x+y-2xy)/(1-xy)$ does not. Equivalently, $R(u)$ itself is not antisymmetric, violating Axiom B'.

(iii) Arctangent. The function $A(u) = (2/\pi)\arctan(\beta u)$ has Abel transform $\phi(x) = \tan(\pi x/2)$, yielding $C(x,y) = (2/\pi)\arctan(\tan(\pi x/2) + \tan(\pi y/2))$. While this composition is well-defined and smooth on the open square $(-1,1)^2$, it cannot be extended smoothly to the closed square $[-1,1]^2$: the second partial derivatives $\partial^2 C/\partial x^2$ and $\partial^2 C/\partial x\,\partial y$ both diverge as $(x,y) \to (1,1)$ (numerically $\sim \epsilon^{-1}$ along $x = y = 1-\epsilon$), violating the C^∞ boundary regularity requirement of Axiom D. This algebraic (ϵ^{-1}) divergence is characteristic of Abel functions with power-law boundary behavior. Logarithmic collars are the admissible boundary class at this face; D' then selects the projective logarithmic collar arctanh from that class.

(iv) Hard saturation. $\min(u,1)$ is not C^∞ at $u = 1$, violating the smoothness requirement of Axiom B. □

Remark 5. *The logistic case (i) is instructive: it appears to be a genuine alternative, but the composition law plus the projective quotient force it to reduce to* tanh*. This illustrates the power of scale coherence together with D': surface-level freedom in the profile choice is reduced to a boundary-collar choice, and the projective quotient fixes the CRM rapidity representative.*

V. THE AXIOMS IN QUANTUM GRAVITY PROGRAMS

We now check how A–D are satisfied – or at least naturally compatible – in each major quantum gravity program when restricted to cosmological observables. The stronger interior form $D^{\pm}$ and the projective normalization D' are tracked separately as open diagnostics.

A. Loop Quantum Gravity / Loop Quantum Cosmology

In Loop Quantum Cosmology (LQC) [22, 23], the modified Friedmann equation reads:

$$H^2 = \frac{8\pi G}{3}\,\rho\left(1 - \frac{\rho}{\rho_c}\right), \tag{14}$$

where $\rho_c \sim \rho_{\rm Pl}$ is a critical density set by the Barbero-Immirzi parameter $\gamma_{\rm BI}$. This has exactly the structure of a saturation equation: the expansion rate is bounded ($H^2 \leq H^2_{\max}$), and the bounce at $\rho = \rho_c$ replaces the classical singularity.

Axiom A: The critical density ρ_c provides finite geometric capacity. *Axiom B:* For $\rho \ll \rho_c$, the standard Friedmann equation is recovered with $c_1 > 0$. The holonomy corrections are cooperative: the departure from GR grows with curvature. *Axiom C:* In the expanding branch ($\dot{a} > 0$), the solution is monotone. *Axiom D:* LQC does not possess a manifest RG composition law in the sense of Axiom D; the effective equations inherit diffeomorphism invariance from full LQG, which provides structural coherence across scales, but a rigorous derivation of the associative composition property (6) from the LQG Hilbert space remains an open problem (hence the "?" in Table I).

Defining $X = 1 - 2\rho/\rho_c$, the LQC Friedmann equation (14) can be rewritten as $H^2 \propto (1 - X^2)$. The factor $(1 - X^2)$ is the same nonlinearity that appears on the right-hand side of the saturation ODE (1); note, however, that the LQC equation governs H^2 while the CRM ODE governs dX/da – the structural analogy is in the saturation mechanism, not a direct identification of the two equations. The CRM parameters k and Φ_0 would map to the LQG minimum area gap Δ (the smallest nonzero eigenvalue of the area operator) and the Barbero-Immirzi parameter $\gamma_{\rm BI}$, though this mapping has not been rigorously derived.

B. Asymptotic Safety

In the asymptotic safety program [8–10], the gravitational effective action has a UV fixed point $g^* > 0$ where the dimensionless Newton's constant and cosmological constant become scale-independent. The RG flow from UV to IR defines the effective gravitational coupling G_k

and cosmological "constant" Λ_k as functions of the RG scale k.

Axiom A: The UV fixed point provides a finite upper bound on the effective coupling. *Axiom B:* The flow away from the fixed point is cooperative: the relevant direction has positive critical exponent [10]. *Axiom C:* The flow is monotone along the relevant direction (by definition of relevance). *Axiom D:* A clarification is needed. The Wilson RG semigroup $R_{k_1} \circ R_{k_2} = R_{k_1 k_2}$ operates on the space of couplings (the "theory space"), not directly on the response function S_T. To connect the two, one must project the RG flow onto the single relevant direction parametrized by g. In the asymptotic safety scenario with a UV fixed point, the flow near the fixed point is controlled by a finite number of relevant directions; in the simplest truncation (Einstein-Hilbert with G_k, Λ_k), there are typically two relevant directions. The single-field assumption of Axiom D corresponds to restricting attention to the dominant relevant direction, under which the RG semigroup on theory space *induces* a composition law on the one-dimensional response $S_T(g)$. This projection is exact when a single relevant direction dominates (as in the cosmological sector), and approximate otherwise – a limitation shared with the single-field restriction of the theorem (see Limitation 4).

Indeed, asymptotic safety is the QG program where Axiom D is most naturally satisfied, because it *is* a renormalization group statement. The asymptotic safety scenario for cosmology [24] produces effective Friedmann equations with saturation behavior at high curvature, consistent with the tanh form.

Recent computations within increasingly sophisticated truncations provide growing numerical evidence for the Reuter fixed point. Eichhorn and Held [7] demonstrate that asymptotic safety constrains the top-quark Yukawa coupling, yielding a prediction of approximately 171 GeV for the top-quark pole mass, consistent with the observed value of 172.6 ± 0.3 GeV [29] – a concrete particle-physics consequence of the UV fixed point. More broadly, Eichhorn [6] reviews how the fixed-point structure naturally accommodates the Standard Model matter content without additional fine-tuning. These results strengthen the case that the asymptotic safety program naturally realizes the axioms: the fixed-point existence (Axiom A), cooperative flow (Axiom B), and RG composition (Axiom D) are properties verified in state-of-the-art functional RG computations.

C. String Theory and Holography

In string theory, the UV finiteness of the string amplitude provides Axiom A. The string landscape, while introducing a selection problem, does not violate the axioms for any *specific* vacuum: within a given compactification, the low-energy EFT has a unique RG flow satisfying Axioms B–D.

The holographic approach (AdS/CFT [14]) provides particularly strong support for Axiom D: the bulk/boundary correspondence *is* a composition law relating geometric descriptions at different scales. The Ryu-Takayanagi formula [25] and its quantum corrections establish a monotone relationship between entanglement entropy and geometric area, satisfying Axiom C.

For cosmological applications, the de Sitter entropy bound $S_{\mathrm{dS}} = \pi/(\Lambda G) < \infty$ provides Axiom A, while the holographic RG flow [26] provides the composition structure.

D. Causal Set Theory

In causal set theory [15, 16], the cosmological constant arises as a dynamical quantity $\Lambda \sim 1/\sqrt{N}$, where N is the number of causal set elements. This produces a cosmological "constant" that decreases as the universe grows, approaching a finite asymptotic value – precisely the saturation behavior of Axiom C.

Axiom A: Finite density of causal set elements provides geometric capacity. *Axiom B:* While the Poisson sprinkling itself is uncorrelated, the *dynamics* of causal set evolution (sequential growth models [16]) produce cooperative ordering: newly added elements preferentially attach to the existing causal structure, enhancing geometric order. *Axiom C:* The $1/\sqrt{N}$ scaling is monotonically decreasing. *Axiom D:* The sequential growth dynamics provide a natural composition law, though its precise relationship to the RG composition of Axiom D requires further analysis.

E. Causal Dynamical Triangulations

Causal Dynamical Triangulations (CDT) [32, 33] construct quantum gravity via a regularized path integral over causal triangulations. Monte Carlo simulations reveal a phase diagram with a physically relevant "de Sitter phase" (phase C_{dS}) in which the emergent geometry has Hausdorff dimension $d_H \approx 4$ and a volume profile well-described by Euclidean de Sitter space.

Axiom A: The triangulation provides a finite number of building blocks per spacetime volume, enforcing finite geometric capacity. *Axiom B:* In the de Sitter phase, the volume profile grows cooperatively before saturating – the geometric ordering is self-reinforcing. The odd-power structure (Axiom B') has not been explicitly verified. *Axiom C:* The de Sitter volume profile is monotone (in the expanding branch). *Axiom D:* CDT possesses natural blocking transformations (coarse-graining of triangulations), providing a candidate for the composition law, but the precise associative structure has not been rigorously established.

F. Noncommutative Geometry

Connes' program [17] reconstructs geometry from spectral data (the Dirac operator on a noncommutative algebra). The spectral action principle yields the Standard Model Lagrangian coupled to gravity from the spectrum of a single operator.

Axiom A: The spectral action has a natural UV cutoff Λ (the energy scale at which the spectral sum is evaluated), providing finite geometric capacity. *Axiom B:* The Seeley-DeWitt coefficients $a_0, a_2, a_4, \ldots$ of the heat kernel expansion have even powers of Λ; the odd-power structure required by Axiom B emerges after reparametrization to a suitable coupling variable, but this identification requires further analysis. *Axiom C:* The spectral flow is monotone in the appropriate direction. *Axiom D:* The composition of spectral triples provides algebraic composition coherence.

G. Summary: Why Universality?

Table I summarizes the axiom verification. No single external QG program has been rigorously proven here to satisfy A–D, $D^{\pm}$, and D' simultaneously; rather, each program naturally satisfies a substantial subset, with the remaining axioms being plausible but not yet formally established (indicated by (✓) and ? in the table). D' is especially important: it is not verified for the listed UV programs in this paper, but is imposed by the CRM effective channel as a projective response normalization. The theorem's value is therefore twofold: (i) as a *conditional* result, it identifies the precise conditions under which tanh saturation is inevitable, and (ii) as a *diagnostic*, it pinpoints which properties of each QG program require further investigation. The broad compatibility across programs explains the Abel/collar-level universality of saturation dynamics; the exact tanh representative additionally requires the projective quotient.

TABLE I. Axiom verification across quantum gravity programs. ✓ = naturally satisfied with explicit construction, (✓) = plausible with mild assumptions, ? = open question requiring further analysis. The D' column records whether the projective boundary quotient is derived or explicitly assumed.

Program	A	B/B'	C	D	D'
LQG/LQC	✓	✓	✓	?	?
Asymptotic safety	✓	(✓)	✓	✓	?
Strings/holography	✓	(✓)	(✓)	(✓)	?
Causal sets	✓	?	✓	?	?
CDT	✓	(✓)	(✓)	?	?
NCG	✓	?	?	?	?
CRM (effective)	✓	✓	✓	✓	assumed

VI. SPACETIME AS AN ORDERED PHASE

The broad axiom satisfaction documented in Table I raises a conceptual question: if the projective tanh representative recurs whenever D' is physically realized, what does this universality *mean* for the nature of spacetime?

A. The conceptual shift

The Saturation Theorem, combined with the CRM's empirical success, suggests a reconceptualization of quantum gravity:

> *Quantum gravity is not the quantization of the metric, but the theory of how geometric order emerges from a pre-geometric quantum system.*

The tanh saturation is not a property of a specific field or coupling, but the universal signature of a capacity-limited cooperative ordering process – independent of whether the microscopic degrees of freedom are spin networks (LQG), strings, causal set elements, or code qubits (QEC).

B. The ferromagnetic analogy

The analogy with spontaneous magnetization is precise at the level of the Ginzburg-Landau free energy. The CRM saturation ODE $dX/da = k(1 - X^2)$ is the equation of motion for the order parameter of a second-order phase transition with double-well free energy $F(X) = -k(X - X^3/3)$. Paper III [3] develops this analogy in detail, identifying:

Ferromagnetism	CRM Cosmology
Magnetization M	Curvature return Ω_Φ
Inverse temperature J/k_BT	Scale factor a
Curie point (J/k_BT_c)	Transition $a_{\rm trans}$
Spin interaction J	Curvature coupling k
Saturation M_s	Saturation Φ_0
$\tanh(J/k_BT)$	$\tanh(k(a - a_{\rm trans}))$

The Saturation Theorem provides the mathematical foundation for this analogy at the *mean-field level*: both systems satisfy an Abel-type composition structure in the mean-field approximation, and the usual mean-field magnetization corresponds to the projective tanh representative. We note that the exact magnetization of a d-dimensional Ising model deviates from the mean-field tanh near the critical temperature ($M \sim (T_c - T)^\beta$ with non-trivial critical exponent $\beta \neq 1/2$ for $d < 4$). This deviation corresponds to the breakdown of the one-dimensional projective-collar approximation near the *critical point* (the onset of ordering), not near *saturation* ($T \to 0$). The analogy is therefore precise for the saturation regime and mean-field theory, while fluctuation corrections near criticality require a multi-field extension (see Outlook). The universality is structural, not metaphorical, within the domain of validity of the axioms and D'.

Having established the ontological framework, we now turn to its empirical consequences.

VII. OBSERVATIONAL CONSEQUENCES AND OPEN QUESTIONS

A. What the theorem predicts

The Saturation Theorem has two classes of predictions:

Projective-collar predictions (after D' is imposed):

1. If the projective boundary quotient is physically realized, the cosmological expansion history should be compatible with tanh-type saturation. We emphasize that the CRM's $\Delta\chi^2 = -5.5$ result [2] does *not* constitute an independent test of the theorem, since the CRM was constructed with a tanh profile before the axioms were formulated. The theorem's empirical content is the *composition law* $\mathcal{C}(x,y) = (x+y)/(1+xy)$, which is a prediction beyond any individual profile fit. An independent test would require either (a) fitting a model-agnostic tanh-template to SN+CMB+BAO data without CRM-specific priors, or (b) verifying the composition law directly by comparing saturation responses at different redshift bins and checking consistency with the velocity-addition rule.
2. The running curvature coupling $\beta_{\mathrm{eff}}(a)$ must be a monotone transition function (already confirmed: $\beta_{\mathrm{early}} \approx 2.8 \to \beta_{\mathrm{late}} \approx 2.0$ [2]).
3. No alternative relaxation profile (logistic, power-law, etc.) can claim theorem-level support unless it also realizes a scale-coherent composition law and explains the projective quotient.

UV-dependent predictions (discriminating between completions):

1. The specific values of k (saturation rate), Φ_0 (saturation amplitude), and γ (coupling constant) depend on the UV theory.
2. The transition sharpness (n in the sigmoidal transition) encodes the universality class.
3. Sub-Planckian corrections to the tanh profile are UV-theory-specific.

B. The missing piece: why *these* parameters?

The Saturation Theorem determines the Abel/collar structure and, under D', the projective *form*, but not the *coefficients*. The remaining open question is:

> *Why does the universe have the specific values $k \approx 9.8$, $\Phi_0 \approx 0.43$, $\gamma \sim \mathcal{O}(1)\, H_0^{-2}$?*

This question is analogous to the fine-structure constant problem: the *existence* of electromagnetic coupling is forced by gauge symmetry, but the *value* $\alpha \approx 1/137$ is not determined by the symmetry alone.

Several UV completions make specific predictions:

- **LQC:** Φ_0 maps to the Barbero-Immirzi parameter γ_{BI}, which is constrained by black hole entropy [27]. This would predict a specific value of Φ_0.
- **Asymptotic safety:** The critical exponents at the UV fixed point determine the transition sharpness n, providing a specific prediction for the shape of $\beta_{\mathrm{eff}}(a)$ [9].
- **Holography:** The de Sitter entropy $S_{\mathrm{dS}} = \pi/(\Lambda G)$ relates Φ_0 to the ratio of UV and IR scales.

A concrete test case is the $\sqrt{\pi}$ conjecture of Paper II [2]: the CRM predicts a MOND-like enhancement factor $\mu_{\mathrm{eff}} = \sqrt{\pi}$ arising from Gaussian normalization of the saturation partition function. Deriving this value from a specific UV completion would directly connect quantum gravity parameters to observed galactic dynamics.

C. Scalar field oscillations as a QG signature

Paper III [3] shows that the Pöschl-Teller potential supports a discrete spectrum of bound states. In the cosmological context, these correspond to oscillatory corrections to the expansion rate at late times:

$$H^2(a) = H^2_{\mathrm{smooth}}(a)\left[1 + \epsilon \cdot e^{-\Gamma a}\cos(\omega a + \delta)\right], \quad (15)$$

with $\epsilon \ll 1$. These oscillations would be a direct signature of the *quantum nature* of the saturation mechanism – analogous to the discrete energy levels of a quantum system. Detection in high-precision BAO or SN data would constitute evidence for the quantum origin of the CRM saturation.

VIII. DISCUSSION AND OUTLOOK

A. What has been achieved

1. **A conditional No-Go theorem:** Any cosmological theory satisfying Axioms A–D, B', $D^{\pm}$, and projective boundary-quotient normalization D' must produce tanh-type saturation. Without D', the same structural hypotheses produce the Abel/collar structure.
2. **Universality clarified:** The observation that LQG, asymptotic safety, strings, causal sets, and NCG all produce structurally similar saturation

dynamics is no longer mysterious at the Abel-structure level; the precise tanh representative additionally depends on the projective boundary quotient.

3. **CRM calibrated:** The CRM's saturation ODE $dX/da = k(1 - X^2)$ is not an ad hoc phenomenological fit, but the canonical projective normal form of capacity-limited cooperative relaxation.

4. **Ontological clarity:** Quantum gravity is not the quantization of the metric, but the theory of geometric order – a phase transition from a pregeometric quantum system to classical spacetime.

B. Relation to the CRM program

This paper completes the conceptual arc of the CRM series: game-theoretic foundation (Paper I [1]), phenomenological fit with $\Delta\chi^2 = -5.5$ (Paper II [2]), Lagrangian derivation (Paper III [3]), galactic dynamics (Paper IV [4]), and now axiomatic foundation (this work). The logical status of the CRM is:

1. The *form* of the dynamics (tanh) is a mathematical theorem once the signed interior composition hypotheses and the projective boundary quotient D' are accepted; without D' the theorem proves the Abel/collar structure.

2. The *Lagrangian* is derived (Paper III).

3. The *parameters* are constrained by data (Papers II, III, IV).

4. The *UV completion* remains open – and must explain why the projective boundary quotient, rather than a different smooth collar, is the physical composition variable.

C. Post-Zookeeper operator transfer

The later FST/Zookeeper proof vocabulary sharpens the role of this paper while sharpening its proof obligations. In that vocabulary, a stability claim is strongest when it can be written as a Galerkin-compatible operator family

$$A_X^{(N)} = H_X^{(N)} + B_X^{(N)}, \qquad \|(I - P_{V_X})k_X\| \le \frac{r_X}{g_X},$$

where $B_X^{(N)}$ is a rank-one or finite-rank defect, r_X is the combined secular/quasimode residual, and g_X is a coercive complement gap. For the present Saturation Theorem the idealised situation is the zero-defect projective limit: the RG composition law is exact, the Abel coordinate is one-dimensional, D' fixes the projective collar, and the hyperbolic composition $\mathcal{C}(x, y) = (x+y)/(1+xy)$ has no off-channel leakage.

The useful transfer is therefore not a new proof of the theorem, but a proof obligation for perturbations of it. Multi-field corrections, stochastic inflation, EFT-of-dark-energy perturbations, or nonperturbative UV–IR matching effects should be treated as defects B_X of the one-dimensional saturation channel. A future extension is strong only if the residual produced by those defects is dominated by a coercive stability gap. In this sense the Zookeeper transfer turns the paper's qualitative caveat—"up to reparametrization and within a single deterministic saturation channel"—into the quantitative test $r_X/g_X \ll 1$.

D. Limitations and caveats

1. **Axiom D and D' are strong.** The renormalization-group composition law is restrictive, and the projective boundary-quotient normalization is an additional mathematical and physical choice. Theories without a well-defined RG flow (some discrete approaches, multiverse scenarios) may not satisfy D; theories with a different boundary collar may satisfy D but fail D'. This distinction is essential: D gives scale coherence, D' selects the tanh representative.

2. **The proof is algebraic, not constructive.** The theorem establishes the Abel structure of the response and, under D', its projective tanh representative, but it does not construct the UV theory. The "reparametrization" $u(g)$ absorbs microscopic information; determining $u(g)$ from first principles requires solving the specific QG theory. A potential objection is that any bounded monotone function can trivially be written as $\tanh(\alpha\, u(g))$ for a suitable $u(g)$, making the profile shape contentless. The non-trivial content therefore resides in the *composition law* and, for the tanh selection, in the projective quotient law $R(C) = R(x)R(y)$.

3. **Cosmological restriction.** The axioms are formulated for cosmological observables. Whether they extend to local physics (black holes, gravitational waves) requires further analysis. The CRM's existing predictions for local physics (chameleon screening, $\alpha_T = 0$, Bullet Cluster resolution [2–4]) are derived from the Lagrangian, not from the axioms.

4. **The continuous group classification and single-field restriction.** The proof relies on Aczél's representation theorem for continuous associative operations [20] and the classification of one-dimensional formal group laws [21]. While these results are well-established, the physical applicability requires the assumption that the composition is *one-dimensional* – i.e., that there is a single relevant direction. This is a genuine limitation: the

CRM itself (Paper IV [4]) introduces a vector sector alongside the scalar, and realistic modified gravity theories (e.g., Horndeski, beyond-Horndeski) generically have multiple coupled degrees of freedom. However, we note that the theorem applies to the *scalar sector in isolation*: the CRM's scalar saturation channel $\Omega_\Phi(a) = \Phi_0 \tanh(k(a - a_{\text{trans}}))$ is a one-dimensional response that satisfies the axioms independently of the vector sector. The multi-field extension (classifying multi-dimensional commutative group laws, including elliptic and Lubin-Tate formal groups) is an open mathematical problem that would strengthen the theorem but is not required for the scalar-sector result.

5. **Boundary-collar freedom.** Smooth boundary behavior alone does not select arctanh. The counter-collar $\phi_\varepsilon = \operatorname{arctanh} + \varepsilon x$ preserves a smooth positive boundary collar but changes the Abel coordinate. The uniqueness selection is therefore deliberately moved to D': once the projective quotient $R = (1 + x)/(1 - x)$ is required to multiply under composition, $\operatorname{arctanh} = \frac{1}{2} \log R$ follows immediately and uniquely up to scale.

E. The No-Go theorem as a filter

The Saturation Theorem serves as a *filter* for quantum gravity proposals:

- Any proposed QG theory that violates Axiom A (allows infinite geometric order) must address the singularity problem separately.
- Any theory that violates Axiom C (has oscillating or multi-fixed-point dynamics) must explain why the observed universe has a unique thermodynamic arrow.
- Any theory that violates Axiom D (lacks scale coherence) must explain why its predictions are insensitive to the observation scale.
- Any theory that satisfies D but violates D' must identify which non-projective boundary collar replaces the CRM rapidity coordinate.

Theories satisfying Axioms A–D together with B' and $D^\pm$ are constrained to an Abel/collar saturation law. Theories satisfying these structural hypotheses and D' are constrained to tanh saturation, providing a concrete, falsifiable prediction for their cosmological sector.

F. Relation to other frameworks

Swampland conjectures. The de Sitter swampland conjecture [30] posits $|\nabla V|/V \geq c \sim \mathcal{O}(1)$ in any quantum gravity landscape, disfavoring a strict cosmological constant. The Saturation Theorem is complementary: it does not address the existence of de Sitter vacua, but constrains the *dynamical approach* to the late-time attractor. A CRM-type saturation with $\Phi_0 < \infty$ is compatible with the swampland bound, since the effective dark energy density $\rho_\Phi(a)$ evolves dynamically rather than being a fixed Λ.

EFT of Dark Energy. The effective field theory of dark energy [31] parametrizes deviations from ΛCDM at the level of the perturbed action. The Saturation Theorem operates at a different level: it constrains the *background* response function, not the perturbative operator basis. A future task is to express the tanh saturation in the EFT-of-DE language, mapping the CRM parameters (k, Φ_0, γ) to the EFT operator coefficients $\{\alpha_i(t)\}$.

G. Outlook

Three directions are immediate:

1. **Parameter derivation.** The next step is to derive k, Φ_0, and γ from a specific UV completion. The most promising candidate is the LQG/holographic intersection, where the Barbero-Immirzi parameter and the holographic entropy bound may jointly fix the saturation parameters.

2. **Multi-field extension.** Extending the theorem to theories with multiple saturation channels (e.g., scalar + vector in the CRM [4]) requires the classification of multi-dimensional commutative group laws (the higher-dimensional analog of the Abel-equation reduction). This is mathematically richer and may yield constraints on the vector sector parameters K_B and $\mathcal{F}$.

3. **Black hole interior.** If the Saturation Theorem applies to black hole interiors (with the radial coordinate replacing the scale factor) and the projective boundary quotient is the correct interior collar, it predicts tanh-type singularity resolution – consistent with LQG bounce models [22] and the Planck star scenario [28].

IX. CONCLUSION

We have proven a corrected Saturation Theorem: finite geometric capacity, cooperative reinforcement, monotone cosmic control, and renormalization-group composition reduce one-dimensional cosmological saturation to an Abel-coordinate boundary-collar structure. The tanh profile follows when the saturation boundary is additionally normalized by the projective quotient $R = (1 + x)/(1 - x)$.

The theorem has three consequences:

1. The CRM's saturation ODE $dX/da = k(1 - X^2)$ is the *projective normal form* of capacity-limited cooperative relaxation – the unique response after fixing the boundary quotient.

2. The observation that LQG, asymptotic safety, string theory, causal sets, and noncommutative geometry all produce similar saturation structures is explained at the Abel/collar level; the specific tanh representative requires the projective quotient.

3. Quantum gravity is reconceptualized as the theory of geometric order – a phase transition from a pregeometric quantum system to classical spacetime – rather than the "quantization of the metric."

The remaining open problem – deriving the specific parameters (k, Φ_0, γ) and the projective boundary quotient from a UV completion – is the analog of deriving the fine-structure constant from a theory of quantum electrodynamics. The theorem constrains the *normal form* of any viable theory once the boundary quotient is fixed.

> *The saturation profile is constrained to the* tanh *type by scale coherence plus the projective boundary quotient; bare smooth boundary regularity gives the wider Abel-collar class.*

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